

# Emmanuel Asinas

Healthcare Insurance Dataset Analysis — BigQuery SQL

Source: <https://www.kaggle.com/datasets/muqaddasejaz/healthcare-insurance-datasets>

**1. What is the average insurance charge by region? Identifies geographic cost disparities — directly relevant to insurance pricing and network planning.**

## INSIGHT

**The top grossing region by Insurance charges is the southeast followed by the northeast.**

```
SELECT ROUND(AVG(charges), 0) AS Avg_Charges, region  
  
FROM projectok-486019.Healthcare.Insurance_Data  
  
GROUP BY region  
  
ORDER BY Avg_Charges DESC;
```

| Query results   |             |           |
|-----------------|-------------|-----------|
| Job information |             | Results   |
| Row             | Avg_Charges | region    |
| 1               | 14735.0     | southeast |
| 2               | 13406.0     | northeast |
| 3               | 12418.0     | northwest |
| 4               | 12347.0     | southwest |

2. What is the average charge for smokers vs non-smokers, broken down by region? Quantifies the smoking cost burden across geographies — a core actuarial question.

### INSIGHT

There are more charges on average for people who smoke than there are charges for people who don't smoke. An average of \$32,050 for smokers compared to \$8,434 who don't smoke.

```
SELECT ROUND(AVG(charges), 0) AS Avg_Charges, smoker
FROM projectok-486019.Healthcare.Insurance_Data
```

```
GROUP BY smoker  
  
ORDER BY Avg_Charges DESC;
```

| Query results   |             |         |        |
|-----------------|-------------|---------|--------|
| Job information |             | Results | Visual |
| Row             | Avg_Charges | smoker  |        |
| 1               | 32050.0     | true    |        |
| 2               | 8434.0      | false   |        |

3. What is the average charge by BMI category (Underweight <18.5,

Normal 18.5–24.9, Overweight 25–29.9, Obese 30+)? BMI bucketing

is standard clinical practice — this tests your ability to use CASE

**WHEN** to create meaningful categories.

### INSIGHT

The average charges by BMI are listed below, the obese classification of the BMI incurs the largest average amount of charges.

## Query results

| Job information |               | Results    | Visualization | JS |
|-----------------|---------------|------------|---------------|----|
| Row             | BMI_Indicator | avg_charge |               |    |
| 1               | Obese         | 15480.0    |               |    |
| 2               | Overweight    | 10994.0    |               |    |
| 3               | Normal        | 10379.0    |               |    |
| 4               | Underweight   | 8852.0     |               |    |

```
SELECT

    BMI_Indicator,

    ROUND(AVG(Rounded_Charge), 0) AS avg_charge

FROM (

    SELECT

        ROUND(charges, 0) AS Rounded_Charge,

        CASE

            WHEN bmi < 18.5 THEN 'Underweight'

            WHEN bmi BETWEEN 18.5 AND 24.9 THEN 'Normal'

            WHEN bmi BETWEEN 25 AND 29.9 THEN 'Overweight'
```

```
ELSE 'Obese'

END AS BMI_Indicator

FROM projectok-486019.Healthcare.Insurance_Data

)

GROUP BY BMI_Indicator

ORDER BY avg_charge DESC;
```

#### 4. Is there a correlation between BMI and insurance charges?

Tests whether higher BMI predicts higher costs — a  
foundational health economics question.

#### INSIGHT

According to the correlation coefficient when comparing the two variables of BMI and charges, the correlation coefficient is around 0.2, it's closer to 0 than it is to 1 so therefore I conclude that there isn't a statistically significant relationship between BMI and charges due to the correlation coefficient being closer to 0 than it is to 1 which would indicate a positive correlation between the two variables.

Query results

| Job information |                   | Result |
|-----------------|-------------------|--------|
| Row             | Correlation       |        |
| 1               | 0.198340968833... |        |

```
SELECT

CORR(charges, bmi) AS Correlation

FROM projectok-486019.Healthcare.Insurance_Data
```

5. Is there a correlation between age and insurance charges? Validates actuarial assumptions about aging and healthcare utilization.

### INSIGHT

Similar to the previous question, the correlation coefficient between someone's age and the amount of insurance charges is closer to 0 than it is to 1, so therefore I can conclude that there is no statistically significant correlation coefficient which proves a correlation between the two variables.

```
SELECT
CORR(age, charges) AS Correlation

FROM projectok-486019.Healthcare.Insurance_Data
```

| Query results   |                   |
|-----------------|-------------------|
| Job information |                   |
| Row             | Correlation       |
| 1               | 0.299008193330... |

6. Who are the top 10% highest-cost patients and what do they have in common? High-cost patient identification is one of the most common real-world healthcare analytics tasks.

### INSIGHT

As can be seen from all of the risk factors which contribute to the top 10% of people being charged the most, the only statistically significant correlation I would find is age because its correlation coefficient between age and charges is closer to 1 meaning there's a higher probability that there is a correlation between the two variables. Bmi is also a factor however it is not as statistically significant as age due to it being 0.317 which is closer to 0 than it is to 1.

| Query results   |             |             |
|-----------------|-------------|-------------|
| Job information |             | Results     |
| Row             | column_name | correlation |
| 1               | age         | 0.5577      |
| 2               | bmi         | 0.3172      |
| 3               | smoker      | 0.1703      |
| 4               | sex         | -0.0582     |
| 5               | region      | -0.0511     |
| 6               | children    | -0.0054     |

```
SELECT column_name, ROUND(correlation, 4) AS correlation
FROM (
  SELECT
    CORR(age, charges) AS age,
    CORR(bmi, charges) AS bmi,
    CORR(CAST(children AS FLOAT64), charges) AS children,
    CORR(IF(smoker = TRUE, 1.0, 0.0), charges) AS smoker,
    CORR(region_rank, charges) AS region,
    CORR(sex_rank, charges) AS sex
```



```

FROM (
  SELECT
    age,
    bmi,
    children,
    smoker,
    charges,
    DENSE_RANK() OVER (ORDER BY region) AS region_rank,
    DENSE_RANK() OVER (ORDER BY sex) AS sex_rank
  FROM projectok-486019.Healthcare.Insurance_Data
  WHERE charges > (
    SELECT PERCENTILE_CONT(charges, 0.99) OVER()
    FROM projectok-486019.Healthcare.Insurance_Data
    LIMIT 1
  )
)
UNPIVOT(correlation FOR column_name IN (age, bmi, children, smoker, region, sex))
ORDER BY ABS(correlation) DESC;

```

| age ▾ | sex ▾  | bmi ▾ | children ▾ | smoker ▾ | region ▾  | charges ▾ | bmi_category ▾ |
|-------|--------|-------|------------|----------|-----------|-----------|----------------|
| 54    | female | 47.4  | 0          | true     | southeast | 63770.0   | Obese          |
| 45    | male   | 30.4  | 0          | true     | southeast | 62593.0   | Obese          |
| 52    | male   | 34.5  | 3          | true     | northwest | 60021.0   | Obese          |
| 31    | female | 38.1  | 1          | true     | northeast | 58571.0   | Obese          |
| 33    | female | 35.5  | 0          | true     | northwest | 55135.0   | Obese          |

```

SELECT
  age,
  sex,

```

```

ROUND(bmi, 1) AS bmi,
children,
smoker,
region,
ROUND(charges, 0) AS charges,
CASE
  WHEN bmi < 18.5 THEN 'Underweight'
  WHEN bmi BETWEEN 18.5 AND 24.9 THEN 'Normal'
  WHEN bmi BETWEEN 25 AND 29.9 THEN 'Overweight'
  ELSE 'Obese'
END AS bmi_category

FROM projectok-486019.Healthcare.Insurance_Data
WHERE charges > (
  SELECT PERCENTILE_CONT(charges, 0.90) OVER()
  FROM projectok-486019.Healthcare.Insurance_Data
  LIMIT 1
)
ORDER BY charges DESC;

```

**Analysis by Emmanuel Asinas · GitHub: [github.com/emmanuelasinas](https://github.com/emmanuelasinas) · BigQuery SQL**